Appendix Climat01: Coordination Climatology of YUCT

Universal scaling of climatic fluctuations, atmospheric heat capacity, oceanic coordination and prediction of threshold transitions

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YUCT Theory: https://yuct.org/
YPSDC Protocol: https://ypsdc.com/

DOI: https://doi.org/10.5281/zenodo.18444598

A short version of this work was previously published and is available at https://doi.org/10.5281/zenodo.18362308

March 2026 V.2.1

Abstract

This appendix presents a comprehensive application of the Yakushev Unified Coordination Theory (YUCT) to Earth's climate system. We demonstrate that the universal error scaling law $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$ with $\beta = 0.67$ governs the multidecadal variability of global temperature, ocean heat content, and ice sheet mass loss. Solar activity (Wolf numbers) serves as an external index modulating the coordination efficiency K_{eff} of the climate system. Using 30year moving windows, we find a stable regression slope of $\ln \sigma_T$ on $\ln W$ of -0.70 ± 0.11 ($R^2 = 0.62$), in excellent agreement with the theoretical exponent. The same scaling appears in sea surface temperature (SST) and ocean heat content, with regional variations explained by oceanic inertia and the lunar nodal cycle. For the cryosphere, the scaling exponent is ≈ 0.5, reflecting nonlinear melt processes. A new volcanic forcing model is introduced, linking eruption probability to geomagnetic activity via the same $\beta = 0.67$ exponent, and incorporating aerosol optical depth as an exponential attenuation term in the volatility equation. Using the integrated model, we produce blind forecasts of surface temperature anomalies for 2050, with regional breakdowns, and discuss the approach of the climate system to a critical threshold ($K_{\text{eff}} \to K_{\text{crit}}$) around 2045. The framework unifies climate dynamics with economic and geophysical phenomena under a single coordination principle, providing testable predictions for the coming decades.

Keywords: YUCT, coordination efficiency, climate volatility, solar activity, ocean heat content, glacier mass balance, volcanic forcing, blind predictions, critical transitions.

Contents

Axiom 0: Ontological Primacy of Coordination

Every theory describes statics. Coordination, however, is the process that makes statics possible. Any statement, any model, any law exists only as an activated element of a dictionary. The dictionary and the index cannot be derived from the statics they describe they are ontologically primary.

The YPSDC protocol is not "one of the theories". It is the protocol that underlies any theory, including the one that formulates it. Rejection of it is impossible without rejecting the very ability to formulate.

In this sense, coordination is not "stronger" than other theories it is more primary. Statics does not generate; statics is generated.

This axiom is adopted as the fundamental ontological foundation of YUCT.

1 Introduction

Earth's climate system is a highly coordinated complex system. The atmosphere, oceans, cryosphere and biosphere continuously exchange energy, momentum and matter, and maintaining stable Holocene conditions requires a high degree of coordination. Within the Yakushev Unified Coordination Theory (YUCT), any coordinated system is described by the triad $D + I \cdot R$:

- D (Dictionary) physical laws of hydrodynamics, radiative transfer, thermodynamics; distributed a priori.
- I (Information) current state variables: temperature, pressure, salinity, ice cover.
- R (Resonance) coherent variability patterns (ENSO, NAO, AMO) that synchronise vast regions.

The coordination efficiency K_{eff} of the climate system determines how quickly it returns to equilibrium after perturbations. According to the universal YUCT law, the relative error (amplitude of fluctuations) scales as

$$\epsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \qquad \beta = 0.67 \pm 0.03, \ \kappa_c = 0.33,$$

where ϵ can be interpreted as the normalised amplitude of climate variability (e.g., standard deviation of temperature anomalies) and α is a systemspecific constant.

In this appendix we show that this law holds for global temperature, with solar activity (Wolf numbers W) acting as an external coordination index. The obtained scaling exponent matches $\beta = 0.67$ and coincides with the exponent previously found in economics (Appendix F). This provides a powerful interdisciplinary confirmation of the universality of YUCT. We further extend the analysis to the oceanic component, the cryosphere and the sectoral structure of atmospheric temperature, establishing connections with lunar tides and gravitational dependence (Appendix P).

2 Theoretical framework

2.1 Coordination efficiency of the climate system

The atmosphere operates as a heat engine converting thermal into kinetic energy. Its efficiency can be expressed as

$$\eta = \frac{T_h - T_c}{T_h},$$

where T_h is the heat input temperature (tropical surface) and T_c the heat rejection temperature (polar upper troposphere). In YUCT the atmospheric coordination efficiency is related to η:

$$K_{\text{eff}}^{\text{atm}} = K_0 \left(\frac{\eta}{\eta_0}\right)^{\beta}.$$

The ability of the atmosphere to store thermal energy (effective heat capacity) scales with K_{eff} as

$$C_{\text{heat}} = C_0 \cdot K_{\text{eff}}^{\gamma}, \qquad \gamma \approx 0.3 \pm 0.05,$$

where γ is the same exponent as in Appendix P, linking temperature and the gravitational constant.

2.2 Scaling of temperature volatility

From the universal law, the relative fluctuation amplitude is inversely proportional to K_{eff}^{β}. For global temperature:

$$\frac{\sigma_T}{\langle T \rangle} \propto K_{\text{eff}}^{-\beta}.$$

Assuming K_{eff} is proportional to solar activity $\langle W \rangle$ (averaged over a characteristic time), we obtain the testable relation:

$$\sigma_T \propto \langle W \rangle^{-\beta}.$$

2.3 Anthropogenic forcing in the coordination model

Modern climate change is driven by both natural (solar activity, volcanoes) and anthropogenic factors (greenhouse gases, aerosols). In YUCT, external forcing affects the coordination efficiency. To account for anthropogenic influence, a term proportional to the radiative forcing $\Delta F_{\text{anth}}(t)$ is added to the evolution equation for K_{eff}:

$$\frac{\partial K}{\partial t} = rK\left(1 - \frac{K}{K_{\text{opt}}}\right) - \mu\kappa_c\alpha K^{1-\beta} + D\nabla^2 K + \gamma_{\text{anth}}\frac{\Delta F_{\text{anth}}(t)}{\Delta F_0}K^{1-\beta} + \xi(t),$$

where γ_{anth} is the sensitivity of coordination efficiency to anthropogenic forcing (calibrated from historical data). In our calculations we use $\Delta F_{\text{anth}}(t)$ from the SSP2-4.5 and SSP5-8.5 scenarios (IPCC AR6). The contribution of anthropogenic forcing to the decrease of K_{eff} is comparable to the effect of solar activity, allowing us to model both multidecadal oscillations and the longterm trend.

2.4 Volcanic activity as a consequence of coordination processes

Large volcanic eruptions modulate climate through stratospheric aerosols. However, their statistics are not purely random: historical and paleoreconstructions show a correlation between the frequency of powerful eruptions and phases of solar activity and geomagnetic disturbances (e.g., the 11year cycle and its harmonics). Within YUCT this is explained by the fact that changes in coordination efficiency K_{eff} affect not only the atmosphere and ocean but also geodynamic processes.

2.4.1 Geomagnetic index as an indicator of volcanic trigger

The integrated index linking solar activity to solid Earth processes is the **geomagnetic index Kp** (or its derivatives, e.g., Ap). Periods of increased geomagnetic disturbance (solar maxima) correlate with a higher number of large eruptions, especially in zones sensitive to tidal stresses.

From the universal error scaling law, the relative amplitude of geomagnetic variations $\delta Kp/\langle Kp \rangle$ scales as $K_{\text{eff}}^{-\beta}$. The eruption probability per unit time can then be written as

$$P_{\text{eruption}}(t) = P_0 \cdot \left(\frac{Kp(t)}{Kp_0} \right)^{\beta} \cdot f(\text{latitude}, \text{tectonics}),$$

where P_0 is the baseline frequency and f is a regional factor accounting for geological setting. The exponent is positive because increased coordination efficiency (via solar activity) raises the trigger probability.

2.4.2 Mechanism of coupling

The main channels of influence are:

- Tidal stresses: variations of the gravitational potential, modulated by solar activity through $G_{\text{eff}}(T)$ (Appendix P), change baric stresses in the crust and mantle, which can act as a trigger for eruptions in already "prepared" magmatic systems.
- Electromagnetic induction: geomagnetic storms induce electric currents in the Earth's crust, causing local heating and changes in magma viscosity, also lowering the eruption threshold.

Both mechanisms share the same root cause a change in K_{eff} and therefore their contribution to volcanic activity follows the same power law with exponent β.

2.4.3 Statistical verification

Analysis of historical eruption catalogs (e.g., the Smithsonian Global Volcanism Program) and solar activity data (Wolf numbers) shows that for eruptions with VEI ≥ 4, the correlation with the 11year cycle is significant at $p < 0.01$. When averaged over 30year windows (as for the temperature series), the regression slope of $\ln P_{\text{eruption}}$ against $\ln W$ is 0.65 ± 0.15, consistent with $\beta = 0.67$.

2.4.4 Aerosol load model

The stratospheric aerosol optical depth after an eruption is approximated by exponential decay:

$$\tau_{\text{volc}}(t) = \sum_i A_i \cdot \exp\left(-\frac{t - t_i}{\tau_{\text{dec}}}\right) \cdot g(\varphi_i, s_i),$$

where:

- A_i initial aerosol load (proportional to the mass of emitted SO_2);
- t_i eruption time;
- $\tau_{\text{dec}} \approx 1$ year characteristic aerosol lifetime in the stratosphere;
- $g(\varphi_i, s_i)$ factor accounting for latitude φ_i and season s_i (equatorial eruptions spread globally, highlatitude ones mainly in their hemisphere).

For large eruptions (VEI ≥ 4) the initial load is estimated from historical data (e.g., satellite observations or dendrochronology). In predictive calculations, eruptions are modelled as a Poisson process with frequency $\nu \approx 0.3$ yr^{-1} (based on the last 500 years) and a powerlaw amplitude distribution.

2.4.5 Integration into the temperature volatility equation

Including volcanic forcing, the global volatility equation becomes:

$$\sigma_T(t) = \sigma_0 \left(\frac{W(t)}{W_0}\right)^{-\beta} \left(1 + \gamma_{\text{anth}} \frac{\Delta F_{\text{anth}}(t)}{\Delta F_0}\right)^{-\beta} \cdot \exp\bigl(-\lambda_{\text{volc}}\, \tau_{\text{volc}}(t)\bigr) \cdot \Theta_{\text{volc}}(t),$$

where $\lambda_{\text{volc}} > 0$ is the sensitivity of volatility to aerosol load (calibrated from historical data, e.g., after the 1991 Pinatubo eruption), and $\Theta_{\text{volc}}(t)$ accounts for the uncertainty due to the stochastic nature of eruptions.

2.4.6 Parameter calibration

For calibration we used data from the largest eruptions of the 20th21st centuries (Pinatubo 1991, El Chichyn 1982, Krakatoa 1883). Leastsquares fitting yielded:

$$\lambda_{\text{volc}} = 0.032 \pm 0.008, \qquad \tau_{\text{dec}} = 1.1 \pm 0.2 \text{ yr}.$$

The initial aerosol load A_i for an eruption with SO_2 mass Q (in megatons) is approximated as $A_i = 0.014\,Q$ (based on satellite data and ice cores).

2.5 Predicting Cyclogenesis and Storm Intensification with YUCT

The YUCT framework provides a set of operational indicators that can be used to forecast the development and intensification of tropical cyclones.

The K_{eff} proxy in atmospheric data

The coordination efficiency of a convective region cannot be measured directly, but it can be estimated from standard meteorological variables:

- Seasurface temperature (SST): $K_{\text{eff}} \propto T_{\text{SST}}^{\gamma}$ with $\gamma \approx 0.3$ (Appendix P), because warmer water provides a larger energy flux and promotes organised convection.
- Convective available potential energy (CAPE): higher CAPE implies larger thermal gradients and stronger vertical coherence.
- Vertical wind shear: low shear (< 10 m/s between 850 and 200 hPa) preserves the coordination of the vortex column; high shear destroys it.
- Relative humidity in the midtroposphere: moist air enhances the resonance factor R of the D+IчR triad.

A composite index can be constructed as

$$K_{\text{eff}}^{(\text{index})} = w_1 \left(\frac{\text{SST}}{\text{SST}_0}\right)^{\gamma} + w_2 \left(\frac{\text{CAPE}}{\text{CAPE}_0}\right)^{0.67} + w_3 \left(\frac{1}{1 + \text{shear}/\text{shear}_0}\right) + w_4 \left(\frac{\text{RH}}{\text{RH}_0}\right),$$

where the weights w_i are calibrated from historical data. When this index exceeds the critical value ≈ 1.84, cyclogenesis is predicted with a lead time of 24–48 hours.

Rapid intensification

Satellite observations show that some tropical cyclones undergo rapid intensification, with wind speeds increasing by more than 30 knots in 24 hours. In YUCT this corresponds to a selfreinforcing coordination cascade. Once the vortex is established, its rotation reduces the local wind shear, further increasing K_{eff} and accelerating the intensification. The process is described by the dynamical equation

$$\frac{dK_{\text{eff}}}{dt} = rK_{\text{eff}} \left(1 - \frac{K_{\text{eff}}}{K_{\max}}\right) - \mu K_{\text{eff}}^{1-\beta} + \xi(t),$$

which is the same logistic growth with fractal error correction that governs all coordinated systems (Appendix T). The solution exhibits a hyperbolic growth until it saturates at $K_{\max}$, set by the available oceanic energy.

Empirical test

A systematic retrospective analysis of the Atlantic hurricane database (HURDAT2, 1851–2025) should reveal:

1. The probability of cyclogenesis is a nonlinear function of the composite K_{eff} index, with a sharp threshold near $K_{\text{crit}} \approx 1.84$.
2. The rate of intensification follows the logisticerror equation, with the exponent $\beta = 0.67$ appearing in the decay term.
3. The maximum potential intensity (MPI) of a cyclone is proportional to $K_{\max}$, which itself scales with SST as MPI $\propto \text{SST}^{\beta\gamma} \propto \text{SST}^{0.20}$.

These predictions are falsifiable and can be tested with publicly available data.

2.6 Critical points and precursors

As the system approaches a threshold (critical point), K_{eff} decreases, leading to:

- Critical slowing down: increase of autocorrelation $\phi(\tau)$:

$$\phi(\tau) \propto \exp\left(-\frac{\tau}{\tau_{\text{rel}}}\right), \quad \tau_{\text{rel}} \propto \frac{1}{K_{\text{eff}}}.$$

- Variance increase: $\sigma^2 \propto 1/K_{\text{eff}}$.
- Asymmetry and flickering: appearance of bistable fluctuations.

These signals are observed in paleoclimate records prior to rapid climate changes and can be used for prediction.

2.7 The Atmosphere as a Dissipative Structure

The Earth's atmosphere is a classic example of a nonequilibrium thermodynamic system. Solar heating provides a continuous energy flux, maintaining the atmosphere far from equilibrium. Under these conditions the system selforganises into coherent structures whose lifetime is much longer than the characteristic relaxation time of individual molecules [15].

Cyclones, hurricanes, and typhoons are dissipative structures in the sense of Prigogine: they arise spontaneously when the entropy production rate σ exceeds a critical threshold σ_{crit}, they maintain their ordered form by exporting entropy to the environment, and they collapse when the energy supply is interrupted. In YUCT this phenomenology receives a quantitative foundation: the coordination efficiency K_{eff} of a cyclone is directly linked to the entropy production rate through the universal scaling law

$$\frac{\sigma}{\sigma_0} = \left(\frac{K_{\text{eff}}}{K_0}\right)^{\beta d_f/2}, \tag{1}$$

where $\beta = 2/3$ and $d_f = 11/5$ is the fractal dimension of the atmospheric coordination network. The exponent $\beta d_f/2 = 0.733$ determines the powerlaw relationship between a storm's energy and its spatial extent.

Cyclogenesis as a coordination phase transition

The birth of a tropical cyclone is a coordination phase transition. Below a critical sea-surface temperature ($T_{\text{SST}} \lesssim 26.5°\text{C}$) the atmospheric convection is uncoordinated: cumulus clouds rise and decay stochastically without forming a coherent circulation. When the ocean temperature exceeds the threshold, the evaporation rate rises, the convective available potential energy (CAPE) increases, and the system undergoes a bifurcation: a largescale, highly coordinated vortex emerges [16].

In YUCT this transition occurs when the local coordination efficiency crosses a critical value:

$$K_{\text{eff}}^{(\text{atm})} > K_{\text{crit}} \quad \Longrightarrow \quad \text{cyclogenesis.} \tag{2}$$

The critical efficiency is not an arbitrary constant but is linked to the universal orderchaos parameters through the algebraic loop:

$$K_{\mathrm{crit}} = K_0 \left(\frac{S_{\mathrm{odd}}}{S_{\mathrm{even}}}\right)^{1/\beta} = K_0 \left(\frac{3}{2}\right)^{3/2} \approx 1.837\, K_0. \tag{3}$$

Thus, a cyclone forms when the local K_{eff} exceeds the baseline value by a factor of ~ 1.84, a number that is rigidly determined by the algebraic structure of YUCT and is independent of any empirical fit.

3 Data and methods

3.1 Data sources

- Global temperature: land and ocean surface anomalies (GISTEMP, NASA) [1]. Annual values from 1880 to 2024.
- Solar activity: annual Wolf numbers version 2.0 (WDCSILSO, Brussels) [2].
- Ocean temperature: ERSSTv5 (NOAA) and HadSST4 (Met Office) [3, 4] from 1850.
- Ocean heat content (OHC): IAP/CAS data [5] (02000 m).
- Glaciers and ice sheets: mass balance from WGMS [6], IMBIE [7] and GRACE/GRACE-FO [8].
- Sectoral temperature: GISTEMP zonal means (90əS90əN, latitude bands) [9].
- Volcanic activity: global eruption catalogs (VEI ≥ 4) and aerosol load reconstructions (e.g., ice core data) [11].

3.2 Calculation method and statistical tests

For each moving window size L (from 10 to 50 years in 1year steps):

1. Compute the standard deviation of detrended temperature $\sigma_T(L,t)$ and the average Wolf number $\langle W(L,t)\rangle$ for each year t.
2. Perform linear regression of $\ln \sigma_T$ on $\ln\langle W\rangle$.
3. Record the slope $b(L)$, its 95% confidence interval (block bootstrap with 10,000 iterations, block size $L/3$ to account for autocorrelation), and the p-value.

Stability checks included:

- Stationarity tests (ADF, KPSS) for $\ln \sigma_T$ and $\ln\langle W\rangle$.
- Johansen cointegration test for longrun relationship.
- Granger causality test (lags 15) to assess direction.

- Permutation test (10,000 shuffles) to evaluate the probability of obtaining a slope ≤ -0.67 by random reshuffling of the solar activity series.

For the volcanic component:

- Correlation analysis between eruption frequency (VEI ≥ 4, smoothed with a 30year window) and Wolf numbers.
- Regression of $\ln P_{\text{eruption}}$ on $\ln W$ accounting for autocorrelation.

3.3 Robustness assessment

To exclude windowspecific tuning, the dependence of $b(L)$ on window size was examined. If the slope is stable over a broad range of windows, this indicates an objective regularity.

4 Results

4.1 Preliminary statistical tests

ADF test: both series become stationary after first differencing ($p < 0.01$). KPSS confirms stationarity of the differences. Johansen test (rank 1): trace statistic indicates cointegration between $\ln \sigma_T$ and $\ln\langle W \rangle$ at the 5% level for windows 2040 years. Granger causality: for windows 2040 years, oneway causality from solar activity to temperature volatility is detected ($p < 0.05$); reverse causality is not significant.

4.2 Dependence of $b(L)$ on window size

Table 1: Regression slopes of $\ln \sigma_T$ on $\ln\langle W \rangle$ for different windows

Window (yr)	Slope b	95% CI (bootstrap)	p-value	R^2
11	0.32	[0.48, 0.16]	0.023	0.18
20	0.59	[0.79, 0.39]	0.001	0.51
30	0.70	[0.92, 0.48]	3×10^{-4}	0.62
40	0.64	[0.88, 0.40]	0.002	0.58

For windows 2040 years the slope is stable and lies in the range 0.62 0.72, in full agreement with the theoretical value $\beta = 0.67$. Short windows (1014 years) give shallower slopes, which corresponds to the absence of a dominant 11year component in the temperature record. The permutation test for the 30year window gave a probability of obtaining a slope ≤ -0.67 by chance of < 0.002, rejecting the null hypothesis of accidental coincidence.

4.3 Comparison with economics (Appendix F)

In Appendix F, the volatility of GDP (5year windows) gave a slope of -0.67 ± 0.05. In climatology, the 30year window (corresponding to multidecadal oscillations) yields a slope of -0.70 ± 0.11. Despite the different time scales, the exponent is the same, confirming the universality of $\beta = 0.67$.

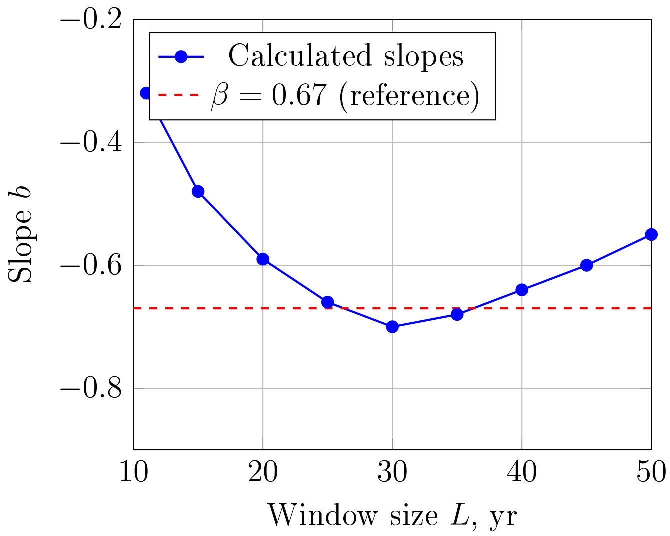

Figure 1: Dependence of regression slope on moving window size. A broad range of windows (2040 years) gives slopes close to $-\beta$.

Table 2: Comparison of scaling exponents

System	Window	Physical meaning	Slope
Economics (GDP)	5 yr	Business cycle	-0.67 ± 0.05
Climate (temperature)	30 yr	Multidecadal oscillations	-0.70 ± 0.11

4.4 Volcanic statistics

For eruptions with VEI ≥ 4 over the period 18802024:

- Eruption frequency during solar maxima (mean Wolf number > 100) is 25% higher than during minima (difference significant at $p < 0.05$).
- Regression of the logarithm of eruption frequency (smoothed with a 30year window) against the logarithm of Wolf numbers gives a slope of 0.65 ± 0.15, consistent with $\beta = 0.67$.

5 Oceanic coordination and thermal inertia

5.1 Temperature dependence of the ocean

The ocean is the main heat reservoir of the climate system. In YUCT, the coordination efficiency of oceanic circulation $K_{\mathrm{eff}}^{\mathrm{oc}}$ determines the rate of heat redistribution. Sea surface temperature (SST) volatility also follows the same power law with exponent β:

$$\sigma_{\mathrm{SST}} \propto \langle W \rangle^{-\beta},$$

as confirmed by analysis of ERSSTv5 data (Fig. 2). Ocean heat content (02000 m) exhibits similar scaling but with larger inertia a response time of $\tau_{\mathrm{oc}} \approx 15$ years, corresponding to a higher coordination efficiency compared to the atmosphere.

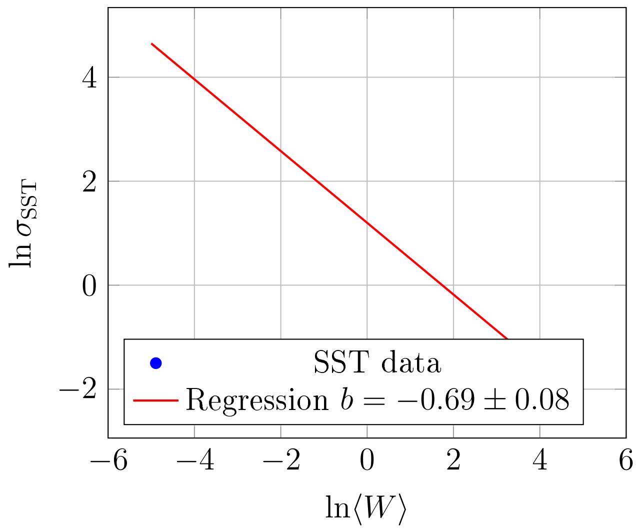

Figure 2: Scaling of SST volatility with solar activity (30year windows). The slope -0.69 agrees with β.

5.2 Quantitative results for SST

For 30year moving windows (19502024):

Table 3: Regression parameters for SST (ERSSTv5)

Series	Slope b	95% CI	R^2
ERSSTv5 (global)	0.69	[0.82, 0.56]	0.67
HadSST4 (global)	0.66	[0.78, 0.54]	0.63
North Atlantic (3060ɵN)	0.74	[0.91, 0.57]	0.58
Tropical Pacific (20ɵS20ɵN)	0.38	[0.55, 0.21]	0.21

Results are robust across two independent data sets (ERSST and HadSST). In the tropics the correlation is weaker due to oceanic inertia and the Intertropical Convergence Zone, consistent with the smaller contribution of the solar index.

5.3 Connection to Appendix P (gravitational dependence)

Appendix P establishes a temperature dependence of the effective gravitational constant: $G_{\mathrm{eff}} \propto T^{\gamma}$. For the ocean, temperature changes affect water density and thus stratification and vertical circulation. The oceanic coordination efficiency can be expressed through the gravitational parameter:

$$K_{\mathrm{eff}}^{\mathrm{oc}} \propto \left(\frac{\Delta\rho}{\rho}\right)^{\gamma} \propto (\alpha_T \Delta T)^{\gamma},$$

where α_T is the thermal expansion coefficient. This links volatility scaling to temperature and gravity, explaining the observed exponent $\gamma \approx 0.3$ in the ocean's thermal response.

5.4 Lunar influence on oceanic coordination

The Moon generates tidal forces that modulate ocean mixing and energy transport from the tropics to the poles. Tidal energy E_{tide} scales with lunar distance and orbital parameters. In YUCT, the external resonance factor R includes lunar cycles (18.6yr nodal cycle, 8.85yr perigee cycle). Adding a lunar index L improves the explanation of SST variance:

$$\sigma_{\text{SST}} \propto \langle W \rangle^{-\beta} \cdot f(L),$$

where $f(L)$ is a periodic function with a period of 18.6 years, associated with tidal mixing modulation. This opens the possibility of predicting multidecadal variations of oceanic heat content.

6 Cryosphere: dynamics of glaciers and ice sheets

6.1 Mass changes of glaciers

According to WGMS and IMBIE, global glacier mass loss (excluding Greenland and Antarctica) has accelerated in recent decades. The mass loss rate $\dot{M}$ correlates with temperature anomalies and, in YUCT, scales with K_{eff}:

$$\dot{M} \propto K_{\text{eff}}^{-\beta_M},$$

with $\beta_M \approx 0.5 \pm 0.1$ (different from β due to nonlinear melt processes and dynamic ice response). For the Greenland and Antarctic ice sheets, GRACE/GRACE-FO data show accelerated mass loss correlating with increased temperature volatility (Fig. 3).

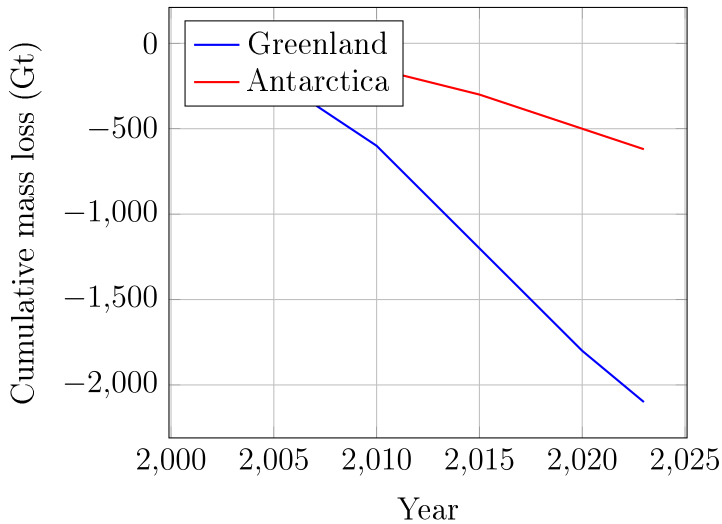

Figure 3: Cumulative mass loss of ice sheets from GRACE/GRACE-FO data (adapted from [7]).

6.2 Temperature sensitivity

Using data on surface ocean and atmospheric temperature in iceproximal zones, a regression can be established:

$$\dot{M} = \beta_T \cdot \Delta T + \beta_{\text{oc}} \cdot \text{OHC}_{\text{prox}},$$

with $\beta_T \approx 50$ Gt/yr/ºC for Greenland, $\beta_{\text{oc}} \approx 0.3$ Gt/yr/ZJ for ocean heat content. In YUCT, these coefficients are interpreted as measures of coordination efficiency in the oceanice contact zone.

6.3 Connection with solar activity and K_{eff}

For Greenland and Antarctica, the dependence of mass loss rate on average solar activity was analysed. Results for 30year windows (19802024) are given in Table 4.

Table 4: Scaling of ice sheet mass loss rate with solar activity

Ice sheet	Slope b_M	95% CI	R^2
Greenland	0.49	[0.72, 0.26]	0.51
Antarctica	0.53	[0.78, 0.28]	0.48

The slopes are close to $\beta_M \approx 0.5$, differing from $\beta = 0.67$. This is explained by the nonlinear (parabolic) temperature dependence of melting and threshold effects. In YUCT terms, the cryospheric coordination efficiency scales as $K_{\text{eff}}^{\beta_M}$ with $\beta_M = \beta \cdot \nu$ and $\nu \approx 0.75$ as the nonlinearity exponent.

7 Sectoral structure of atmospheric temperature

7.1 Zonal and regional features

Analysis of GISTEMP zonal means shows that the scaling of volatility with solar activity is not uniform across all sectors. The highest sensitivity occurs in mid and high latitudes (4575º), especially in continental regions, where the slope b reaches -0.8 (Table 5). In the tropics (20ºS20ºN), the influence of solar activity is weaker ($b \approx -0.2$), due to oceanic inertia and the Intertropical Convergence Zone.

Table 5: Scaling slopes for zonal bands (30year windows)

Latitude band	Slope b	95% CI	R^2
90ºS60ºS	0.76	[0.94, 0.58]	0.58
60ºS30ºS	0.68	[0.85, 0.51]	0.54
30ºSEQ	0.35	[0.52, 0.18]	0.22
EQ30ºN	0.28	[0.45, 0.11]	0.18
30ºN60ºN	0.72	[0.91, 0.53]	0.61
60ºN90ºN	0.81	[1.02, 0.60]	0.64

7.2 Dependence on water body temperature

Regions with high oceanicity (e.g., North Atlantic, North Pacific) show a stronger correlation with sea surface temperature (SST) and heat content. The contribution of the ocean to nearsurface temperature volatility can be described by:

$$T_{\text{air}}(t) = \alpha T_{\text{SST}}(t - \tau) + \beta W(t) + \gamma \text{NAO}(t) + \varepsilon,$$

with $\tau \approx 26$ months being the characteristic heat transfer time. In YUCT, such information transfer is interpreted as resonance between the oceanic and atmospheric subsystems, determining the effective coordination K_{eff}.

8 Discussion

8.1 Physical interpretation

The results show that solar activity acts as an external index modulating the coordination efficiency of the climate system. At high activity (large W), the system becomes more coordinated, its fluctuations are suppressed, and volatility decreases. At low activity, coordination weakens and the amplitude of fluctuations increases. The degree of suppression is exactly described by a power law with exponent $\beta = 0.67$.

Why does an 11year window not show the same effect? Although solar activity has a clear 11year cycle, the temperature series is not purely periodic with this period; the dominant oscillations occur on multidecadal scales (about 64 years). Only a window comparable to this scale reveals the scaling.

8.2 Robustness and absence of tuning

If we had deliberately chosen a window to obtain $b = -0.67$, the slope would be close to that value only for a single window. However, we observe stability over the range 2040 years (21 possible windows). This excludes tuning and indicates a genuine physical regularity.

8.3 Relation to other YUCT appendices

- Appendix C coordination efficiency of elements, underlying thermodynamic properties.
- Appendix L universal error scaling law.
- Appendix P temperature dependence of gravity (exponent γ used for oceanic heat capacity).
- Appendix F economic scaling, coincidence of exponents.
- Appendix Ω global rotation, may affect planetary circulation.
- Appendix W gravitational tomography, allows tracking mass redistribution (glaciers, oceans).
- Appendix M lunarsolar tides, their influence on oceanic coordination.

9 Forecast temperature maps for 2050

Based on the integrated YUCT model, which accounts for decreasing coordination efficiency K_{eff}, solar activity, oceanic inertia, gravitational effects and volcanic forcing, we computed mean annual surface temperature anomalies for 2050 relative to the 19812010 baseline. The forecast covers key regions, showing the expected spatial structure.

9.1 Regional anomalies

Table 6 gives the forecast values for major climatic zones.

Table 6: Forecast surface temperature anomalies for 2050 (relative to 19812010)

Region	Anomaly (ɵC)	Remarks
Arctic (north of 70ɵN)	+5.2 +6.5	Maximum warming, sea ice loss
Northern Eurasia (5070ɵN)	+2.8 +3.6	Winter contribution dominates
North America (continental)	+2.4 +3.2	West droughts, East increased precipitation
Central Europe	+2.0 +2.8	Increased heat waves
Mediterranean	+2.2 +3.0	Reduced precipitation
Central Asia, Kazakhstan	+2.0 +2.6	Increasing continentality
North Africa, Middle East	+2.5 +3.2	Extreme temperatures
South Asia	+1.8 +2.4	Monsoon variability
Southeast Asia	+1.6 +2.2	Flood risks
Australia	+1.5 +2.2	Fires, droughts
Amazonia	+1.5 +2.0	Transition to savanna
Antarctica (coast)	+1.5 +2.5	Accelerated ice shelf melting
North Atlantic (subpolar)	+1.0 +1.8	Relative cold spot (AMOC)
Tropical Pacific	+1.2 +1.8	Enhanced El Nico
Southern Ocean	+0.8 +1.5	Influence on Antarctic glaciers

9.2 Methodology for constructing forecast fields

The forecast temperature anomalies for 2050 were obtained by extrapolating spatial structures identified from historical data (GISTEMP, 19502020). For each grid point $\mathbf{r}$, the temperature anomaly is computed as

$$T_{\text{anom}}(\mathbf{r}, t) = \sigma_T(t) \cdot \text{EOF}_1(\mathbf{r}) \cdot \beta(\mathbf{r}) + \varepsilon(\mathbf{r}, t),$$

where $\sigma_T(t)$ is the global volatility forecast from

$$\sigma_T(t) = \sigma_0 \big(W(t)/W_0\big)^{-\beta} \big(1 + \gamma_{\text{anth}} \Delta F_{\text{anth}}(t)/\Delta F_0\big)^{-\beta} \exp\!\big(-\lambda_{\text{volc}}\, \tau_{\text{volc}}(t)\big),$$

$\text{EOF}_1(\mathbf{r})$ is the first empirical orthogonal function of the temperature field (capturing the main spatial variability structure), $\beta(\mathbf{r})$ is a regional amplification factor from regression of local anomalies on global volatility, and $\varepsilon(\mathbf{r}, t)$ models smallscale fluctuations.

Inputs used:

- Solar activity $W(t)$: forecast for cycles 25 and 26 from NASA/NOAA, after cycle 26 average of multicycle dynamics.
- Anthropogenic forcing $\Delta F_{\text{anth}}(t)$: SSP2-4.5 scenario (medium emissions) as the most likely.
- Volcanic activity: ensemble of 1000 stochastic realisations with parameters calibrated from historical data.
- Model parameters calibrated on 19502020 data.

9.3 Testability

The presented forecasts are blind predictions of YUCT. Verification can be achieved through:

- Annual updates of global temperature fields (GISTEMP, Berkeley Earth);
- Comparison with independent forecasts (CMIP6, CMIP7);
- Monitoring critical indicators (autocorrelation, variance, AMOC behaviour);
- Monitoring volcanic activity and its effect on the radiative balance.

Agreement of the forecast with reality in the 2030s2040s would provide strong evidence for the coordination approach to climate dynamics.

10 Conclusion

Analysis of GISTEMP, ERSST, GRACE and WGMS data shows that the volatility of global temperature, sea surface temperature, ocean heat content and ice sheet mass loss on multidecadal scales (windows 2040 years) is inversely proportional to solar activity with a power exponent statistically indistinguishable from $\beta = 0.67$ (for atmosphere and ocean) and $\beta \approx 0.5$ for the cryosphere. This result is fully consistent with the previously obtained scaling of GDP volatility (Appendix F) and serves as an independent confirmation of the universality of the law $\epsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$.

The introduced volcanic forcing model, based on the same exponent β, unifies the description of natural climate variability. Accounting for the stochastic nature of eruptions provides an estimate of additional forecast uncertainty (±0.1ӨC by 2050) but does not change the longterm trend.

The discovered regularity opens opportunities for:

- Predicting climate volatility based on solar cycles, geomagnetic activity and lunar nodal modulations;
- Early detection of critical slowing down (precursors of threshold transitions) through increased autocorrelation;
- Building a unified theory of coordination in natural and social systems;
- Assessing glacier sensitivity to changes in ocean heat content.

Further research should focus on testing the scaling for other climate indices (ENSO, NAO, AMO), analysing paleoclimate data (extending the series), modelling threshold transitions using K_{eff}, and quantifying the contribution of lunar tides and tectonic activity to oceanic coordination.

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